

Crowdsourcing methodological comparisons: A many-analysts simulation study to identify conditions under which results in convenience samples generalize to target populations

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Abstract

Research on the replicability crisis in empirical psychology has shown that researcher degrees of freedom (RDF) are a large-scale problem. Many-analysts studies, in which independent research teams analyze the same data to address the same questions, have been used to reveal how analytical choices shape substantive conclusions, including, but not limited to, documenting the impact of RDF and exploring ways to mitigate them.

In this paper, we investigate whether similar problems exist in the methodological literature. Specifically, we examine the extent to which RDF in methodological studies may affect conclusions about method performance. Using a many-analysts simulation study, we demonstrate how different analysts, faced with the same underlying research question, may arrive at varying results depending on these decisions. Our case study focuses on the extent to which results obtained in convenience samples generalize to target populations.

We quantify between-analysts variability in methodological conclusions and evaluate whether many-analysts approaches can help identify and reduce overoptimistic claims in method evaluations and improve the replicability of simulation-based evaluations, meaning whether independent researchers addressing the same methodological question arrive at comparable conclusions. We conclude with practical recommendations for designing and organizing many-analysts simulation studies as a feasible tool to improve the robustness and replicability of methodological research.

Keywords: Many-analysts study, researcher degrees of freedom, method comparison, replicability, neutral comparison study

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Researcher degrees of freedom (RDF), referring to the flexibility researchers have in making analytical and methodological choices, and the possibility of (unintentionally) exploiting these RDF to obtain desired results, have been identified as one factor contributing to the replication crisis in science (Simmons et al., 2011).

Although improved statistical methodology has been proposed as one way to increase replicability (Frias-Navarro et al., 2020; Sauerbrei et al., 2014), methodological research itself is not immune to problems arising from RDF and incentives encouraging result-dependent selective reporting (Nießl et al., 2024; Buchka et al., 2021).

One core goal of methodological research is to develop new methods. Since authors are often expected to demonstrate that a newly proposed method outperforms existing methods (Boulesteix et al., 2013; Boulesteix, Binder, et al., 2017), they may be tempted to employ questionable research practices including selective reporting of conditions, methods, or performance measures (Bouma et al., 2026) that may yield overly optimistic results.

Buchka et al. (2021) found that new methods outperformed existing ones in 94.4% of comparisons in papers proposing the new method, compared to 64.4% in neutral evaluations in bioinformatics. Even in the absence of problematic intentions, deeper expertise with a method will often make fair comparisons challenging.

Proposed solutions to improve the trustworthiness of methodological results include neutral comparison studies (Boulesteix et al., 2013; Boulesteix, Wilson, & Hapfelmeier, 2017), preregistration and guidelines for the design and reporting of simulation studies (Paxton et al., 2001; Morris et al., 2019; Siepe et al., 2024; Strobl et al., 2026), replication of simulation studies (Lohmann et al., 2022; Luijken et al., 2024), so-called living synthetic benchmarks (Bartoš et al., 2025), and abandoning the “one method fits all data sets” philosophy (Strobl & Leisch, 2024).

Researcher degrees of freedom in simulation studies

Simulation studies are an important tool for evaluating and comparing statistical methods. They are widely used to study methodological properties and to guide applied analysis in statistics, psychometrics, and machine learning. Simulation studies require many design, analysis, and reporting decisions, including choices regarding the data-generating process (DGP), performance measures, method implementation, parameter settings, aggregation, presentation and interpretation of results. As in empirical research, such flexibility introduces RDF that may influence study conclusions.

Methodological comparison studies can therefore easily yield overly optimistic results and may be difficult for independent researchers to replicate, for example when particular DGPs, variants of the same method, or performance measures are selected that favour a particular method (Jelizarow et al., 2010; Nießl et al., 2022; Ullmann et al., 2023; Pawel et al., 2024). Here, replicability refers to the extent to which independent researchers reach comparable conclusions when investigating the same methodological question using independently developed simulation designs.

Even in the absence of result-dependent selective reporting, traditional single-analyst studies typically address broad methodological research questions from the perspective of a single research team, which captures one expert-informed, though non-exhaustive, set of methodological and analytical decisions.

Extending many-analysts studies to simulation-based methodological research

Many-analysts studies assess analytic robustness by inviting multiple independent research teams to address the same research question using the same data (Silberzahn et al., 2018; Aczel et al., 2021, 2026). Rather than treating one analysis as definitive, they quantify how conclusions vary across reasonable analytical choices.

In this paper, we explore whether this framework can be extended to simulation-based methodological research. In this setting, analysts can share the same research question and target estimands while being given freedom in their choices regarding the design, analysis and reporting

of a simulation study. For example, the lead team may fix the research question, the competing methods, and the analysis targets, while analysts independently specify the DGP and relevant implementation choices. In this way, a many-analysts simulation study may provide a pluralistic perspective on broad methodological questions and may help assess the sensitivity of conclusions to alternative, yet plausible, design choices (Aczel et al., 2021; Sarafoglou et al., 2025, 2024).

If analysts with different levels of expertise in the methods under study are included, many-analysts simulation studies may additionally allow researchers to distinguish method performance under expert versus non-expert use as different estimands (Nießl et al., 2024). This distinction is rarely evaluated in traditional single-analyst comparisons, where results implicitly reflect the expertise of a single research team. However, it is often highly relevant for applied readers – especially for complex methods requiring tuning or other implementation decisions.

Even for methodological research questions for which neither incentives to demonstrate the superiority of a preferred method nor differences in expertise are expected, there are often many possible variants of the same method, which can lead to variability in results (Sarstedt et al., 2024). Many-analysts studies may therefore offer a valuable tool to assess variation in results across alternative choices in the design and analysis of simulation studies.

Structuring many-analysts simulation studies

Running many-analysts simulation studies poses many challenges: Simulation studies can be computationally intensive, and even the results of a single simulation study may already be difficult to analyze and communicate clearly. Having each analyst independently run an entire simulation study may therefore generate large volumes of heterogeneous results that are difficult to compare and aggregate.

To facilitate a balance between granting analysts enough freedom in their choices and ensuring that the results remain comparable and amenable to subsequent aggregation, we propose distinguishing and preregistering confirmatory and exploratory elements in many-analysts simulation studies (Lange et al., 2025) to distinguish RDF in the design, analysis, and interpretation.

In a confirmatory part, the lead team can formulate precise hypotheses and restrict analysts' choices to a predefined set of relevant RDF. To reduce errors and time burden, the lead team may provide pre-specified code in which analysts modify only the relevant parts (e.g., the DGP, parameter values varied across scenarios, or method implementations including tuning procedures). Submitted code can be reviewed before being integrated into a single framework executed by the lead team.

The confirmatory part can be complemented by an exploratory part, in which analysts are asked to investigate additional aspects of the research question and consider a broader set of RDF. These may include elements the lead team did not anticipate or had no specific hypotheses about. Importantly, this may be an important contribution of many-analysts simulation studies, as it allows diverse analysts to surface new insights, challenge assumptions, and add perspectives that beyond the confirmatory part. The findings from the exploratory part can be summarized by the analysts in short reports included in the supplementary material, with a brief overall description in the main paper.

Across both parts, clear instructions and structured documentation, such as preregistration and reporting templates pre-filled by the lead team, may help distinguish choices fixed by the lead team from those left open to analysts (Siepe et al., 2024). In light of evidence that researchers' prior expectations may influence analytical decisions (Borjas & Breznau, 2026), it is interesting to collect analysts' subjective assessments through pre- and post-study questionnaires (Sarafoglou et al., 2024). These may also capture potential methodological concerns and provide context for interpreting the analysts' results.

Note that this workflow represents a first proposal on how to conduct many-analysts simulation studies, rather than an established or standardized practice.

In this paper, we present a many-analysts simulation study comparing different effect size measures with respect to their robustness to non-representative sampling mechanisms. Beyond addressing this methodological question, our study serves as an illustrative case study for several meta-scientific research questions. Specifically, we examine (i) the extent to which researchers

exercise different RDF when addressing the same methodological research question, (ii) whether these RDF lead to meaningful differences in results and contribute to heterogeneity in findings, and (iii) how such choices influence the overall conclusions within the specific setting. Furthermore, we outline how many-analysts simulation studies can be conducted, synthesized, and communicated in a transparent and informative manner.

Case study: Identifying conditions under which the results in convenience samples generalize to the target population

A large part of psychological research relies on convenience samples (Andrade, 2021; Wild et al., 2022; Peterson & Merunka, 2014). As a consequence, how well results obtained in non-representative samples generalize to the target population represents an important methodological question in psychology (Yarkoni, 2022).

There is little guidance on the conditions under which results based on convenience samples generalize to the target population, despite their ubiquity and potential to undermine research findings. This relates to longstanding but unresolved debates in the field, in which some argue that representative samples are of utmost importance (Henrich et al., 2010; Smith, 2020; Stratton, 2021). In contrast, others insist that convenience samples are well-suited for fundamental research (Rothman et al., 2013; Sherman, 2024).

Obtaining a full range of values of a psychological construct of interest is often challenging for empirical researchers. For example, individuals with very low impulse control, severe cognitive impairment, or high social anxiety may be unreachable or unable to participate in research. Psychology's reliance on convenience samples (Peterson & Merunka, 2014; Andrade, 2021; Wild et al., 2022) such as student populations and self-selection into empirical studies (Każmierczak et al., 2023; Haynes & Robinson, 2019; Didelez et al., 2010) further exacerbate this problem.

We compared five effect size measures with respect to their robustness to non-representative sampling mechanisms: the unstandardized regression coefficient (β_1), the standardized regression coefficient (β_1^{std}), the coefficient of determination (R^2), Pearson

correlation (r), and Cohen's d (d).

Previous work suggests that restricting the range of predictors often biases correlation coefficients and standardized effect sizes, but not the unstandardized regression coefficient, whereas restricting the range of the outcome may bias all five effect size measures (Sackett et al., 2007; Carretta & Ree, 2022; Le et al., 2016; Bobko et al., 2001; Li, 2015). We further expected restriction in either variable to reduce statistical power because reducing variance generally increases standard errors in linear regression, even when effect estimates remain unbiased.

Assuming a correctly specified multiple linear model, we formulated the following three hypotheses:

- **Hypothesis 1:** When the sampling mechanism restricts the range or distribution of the predictor X , the unstandardized regression coefficient β_1 and Cohen's d remain unbiased, whereas β_1^{std} , R^2 and r are in general biased.
- **Hypothesis 2:** When the sampling mechanism restricts the range or distribution of the outcome Y , all five effect size measures are in general biased.
- **Hypothesis 3:** Restricting the range or distribution of the predictor X or the outcome Y will lead to a loss of power.

In line with these hypotheses, range restriction is expected to affect both what Calder et al. (1981) calls effect application research and theory application research, as it influences both the estimation of the magnitude of effect sizes and their detectability.

Although theoretical results (Supplement S1) motivate these hypotheses, simulation-based evaluations are valuable for assessing the magnitude and practical relevance of bias. The latter may strongly depend on how the DGP is specified. Analysts may reasonably differ in their choices regarding the form and strength of range restriction, which may contribute to variability in conclusions. For example, range restriction may be implemented deterministically through fixed cutoffs (e.g., excluding observations below a threshold) or probabilistically by assigning selection probabilities conditional on the value of X or Y .

Figure 1 illustrates the behaviour of the unstandardized regression coefficient for deterministic and probabilistic restrictions of the range of X and Y.

Moreover, theoretical results may not hold when modelling assumptions are violated, for example due to measurement error or model misspecification. Such considerations may lead analysts to adopt different simulation designs, and may further contribute to heterogeneity in conclusions.

Against this background, we examined how analyst-specific DGP choices influence results in a confirmatory part of our many-analysts study, in which analysts generated data under correctly specified model assumptions. In an exploratory part, analysts were invited to further investigate the hypotheses in situations where the model assumptions are not met.

We neither adjusted for multiple testing in the confirmatory part nor in the exploratory part since we report and interpret the findings of all performed tests on equal footing, irrespective of their significance (Boulesteix & Hoffmann, 2024).

Disclosures

Preregistration

We preregistered all code and materials used for the confirmatory part of the study on OSF (<https://osf.io/4uwae>).

Data and materials

All materials provided to the analysts, including the recruitment text, information letter, questionnaires, instructions, scripts, and helper code, as well as all analysis code in anonymized form for the aggregated results and the analysts' contributions to the confirmatory and the exploratory part are available on the OSF.

Reporting

We report all simulation results contributed by the analysts and summarise the analytical procedures as specified in the analysts instructions, with deviations and analyst-specific details documented in the supplementary materials.

Ethical approval

Ethical approval was obtained from the Ethics Committee of the Faculty of Mathematics, Informatics and Statistics at Ludwig Maximilian University of Munich.

Pilot study

Before we conducted the many-analysts simulation study, we ran a sequential pilot phase ($N = 10$) among co-authors and colleagues to test and refine the study materials and the analysis code, including the instructions, wrapper functions and overall data generation, estimation and aggregation. Results from the pilot study are included in the Supplement S3.

Method

Analysts

We recruited a convenience sample of analysts *via* social media (BlueSky, LinkedIn) and direct email outreach to academic methodological research groups, with a target of 15–50 analysts.

Eligible analysts were required to be at least Master’s-level students with experience in simulation-based research and R. Co-authorship was offered to analysts who made a substantive scientific contribution to the study. $N = XX$ analysts expressed interest and received the study materials, and $N = XX$ submitted a completed analysis script.

Materials

Analysts participated in a confirmatory part addressing the pre-specified hypotheses, and an exploratory part allowing greater flexibility in the design and analysis of the simulation study. They were provided with instructions outlining the aim and background of the study, the timeline, a structured R script template, and ADEMP (Aims, DGP, Estimands, Methods, Performance measures) templates for both the confirmatory part and the overall many-analysts study (Siepe et al., 2024). In addition, we asked them to complete pre- and post-analysis questionnaires on research hypotheses and study design that were adapted from Sarafoglou et al. (2024).

In the confirmatory part, the analysts were instructed to consider only cases where the true data-generating model was a correctly specified linear model and only needed to complete

designated sections related to the DGP. Specifically, analysts were asked to define (i) the DGP, including the distribution of X, any additional covariates, and the linear form linking X and Y; (ii) three effect size scenarios (weak, moderate, strong), defined relative to their own DGP specification; (iii) four levels of range restriction in X and Y (none, weak, moderate, strong); and (iv) the sample size. This structure allowed analysts to address the same overarching hypotheses while varying key DGP and restriction-mechanism choices. The full analyst instructions, templates, and helper scripts are available on OSF.

Analysts were provided with pre-written simulation and analysis code for the confirmatory part. Effect estimates, bias measures, and power-related quantities were computed for each scenario using the true values implied by each analyst's unrestricted DGP. True values were derived analytically or via Monte Carlo simulation, as detailed in Supplement S1. To ensure reproducibility and comparability, analysts used standardized helper scripts and submitted fully reproducible code.

We did not pre-register the exploratory analyses, as the RDF and analytical decisions could not be fully characterized beforehand. Instead, we will descriptively summarize analysts' RDF and collect short reports of their analysis. Where exploratory choices could be systematically characterized and performance measures such as bias or power could be derived, we examined them using meta-regression and descriptive statistics.

Aggregating results and exploring sources of variability

Analysts submitted their completed scripts and pre- and post-simulation questionnaires. All code was independently reviewed by two reviewers, for compliance with the study requirements, successful execution, and for inclusion. We defined a priori data-quality checks and rules for handling method failures to ensure valid simulation results; details are provided in Supplement S1 (Wünsch et al., 2025; Pawel et al., 2025).

The two reviewers also extracted key DGP characteristics, including predictor type, distribution type, use of covariates, presence of covariates correlated with X, effect size strength in terms of the true partial coefficient of determination (R^2), type of range restriction, restriction

strength in terms of variance reduction, and convenience-sample size. Valid submissions were aggregated into a combined simulation study (see Table 1). We descriptively summarize analysts' design choices to characterize the extent of RDF.

Confirmatory evaluation was based on the mean relative bias aggregated across DGPs and its 95% confidence intervals (CIs), stratified by restriction strength and effect size strength. This allows us to examine whether analyst-specific choices contribute to heterogeneity in methodological conclusions across the pre-specified hypotheses. The 95% CIs were compared to a $\pm 5\%$ equivalence region, and effect sizes were classified as unbiased, biased, or inconclusive depending on whether the interval lay entirely within, entirely outside, or overlapped $[-0.05, 0.05]$. Additionally, we investigated sources of variability across analysts and DGPs using descriptive statistics and pre-specified meta-regression models incorporating DGP-related RDF and analysts' prior beliefs from the pre-analysis questionnaire.

Meta-regression models included random intercepts for DGP to account for repeated measures, with an additional analyst intercept when multiple DGPs were contributed. For the RDF analysis, two specifications were estimated: one using subjective effect sizes and restriction levels, and one using objective measures (true partial R^2 and variance reduction). Additional details are provided in Supplement S1.

Results

Analysts characteristics

Of the $N = XX$ submissions, $N = XX$ met the pre-specified inclusion criteria and were included in the analysis (Supplement S2). Most participants were [PhD students] ($XX\%$), primarily from [psychology] ($XX\%$). Overall, analysts reported [moderate-to-high] experience with simulation studies and familiarity with convenience samples; detailed descriptives are provided in Supplement S2.

Exploring researcher degrees of freedom

To address how much researchers differ in their analytical and design choices when addressing the same methodological research question, Table 1 and Figures 2 and 3 summarize

the analyst-specific design choices, subjective design categories, and restriction mechanisms used in the confirmatory part. Together, these results show [substantial] variation in the design choices made by analysts in our case study.

Subjective design categories often mapped onto overlapping objective properties across analysts (Figure 2).

Restriction mechanisms also differed considerably across analysts (Figure 3).

Aggregated methodological conclusions

To examine whether analyst-specific RDF were associated with differences in methodological conclusions, we aggregated simulation results across analysts and compared the resulting conclusions across the pre-specified hypotheses (Figures S2, S3, and S4). Across analysts, the unstandardized regression coefficient was generally classified as [unbiased] under restriction in X, whereas Cohen's d yielded more heterogeneous results and was often classified as inconclusive. Under restriction in Y, all estimators showed bias, consistent with Hypothesis 2 (see S2, S3 and S4).

Figure 4 illustrates the relative bias estimates contributed by individual analysts across DGPs. While some estimators showed relatively consistent behaviour across analyst-specified scenarios, others varied substantially depending on DGP and restriction choices. Similar patterns were observed for Pearson's r and R^2 , which are closely related to the standardized regression coefficient (Supplement S2).

Figure 5 shows power for the unstandardized regression coefficient in restricted and unrestricted samples at a fixed sample size.

Figure 6 illustrates analysts' prior and posterior belief ratings for each hypothesis, with connecting lines indicating within-analyst changes. On average, beliefs shifted toward [greater/smaller] plausibility of Hypothesis 1.1 and 1.2, whereas changes for Hypotheses 1.3, 2 and 3 were [comparatively smaller/larger/similar]. However, substantial heterogeneity in posterior beliefs [remained] across analysts.

Consensus and disagreement across DGPs

To further examine how analyst-specific choices shape methodological conclusions within this setting, we classified individual DGP-scenario combinations using Monte Carlo CIs within each simulation scenario and the same $\pm 5\%$ equivalence rule as in the aggregated confirmatory analysis. We then computed the proportion of DGP-level classifications supporting, contradicting, or remaining inconclusive for each hypothesis.

While some hypotheses showed [substantial] consensus across analysts, others depended strongly on analyst-specific DGP and restriction choices. Supplement S2 further provides results stratified by RDF, demographics, and plausibility.

Sources of heterogeneity in methodological conclusions

To address which analyst-specific choices are associated with differences in methodological conclusions, we examined whether variability in relative bias was related to RDF, analyst characteristics, and prior beliefs using pre-specified meta-regression models (Tables 2, S6, and S7). Meta-regression analyses suggested that several analyst-specified RDF contributed [meaningfully] to heterogeneity in simulation results (Tables 2 and S6).

Tables 2 and S6 report the results of the meta-regression models modelling relative bias as a function of RDF. The two meta-regression specifications yielded [largely consistent/inconsistent] results, suggesting that the observed effects primarily reflect the underlying statistical properties of the simulated data rather than the subjective design categories. The estimated variance components are summarized in Table S5. Across effect size measures, τ^2_{analyst} indicates [large/moderate/small] between-analyst variability in the resulting estimates, while τ^2_{DGP} captures remaining heterogeneity across analyst-specified DGPs.

Table S7 reports the corresponding meta-regression results for analysts' prior beliefs. We observe [no/minor/major] differences in relative bias associated with variation in analysts' prior beliefs.

Overall, the results indicate that differences in analyst-specified simulation designs translated into meaningful heterogeneity in methodological conclusions.

Exploratory part

The exploratory part substantially expanded the range of analyst-specified RDF and simulation settings beyond the constrained confirmatory framework (also see S8). Analysts investigated alternative distributions, sampling mechanisms, estimands, and violations of modelling assumptions that were not captured by the preregistered confirmatory scenarios.

Where possible, we additionally derived relative bias and power estimates for exploratory scenarios; corresponding results are reported in Supplement S2 and the accompanying Shiny application.

From a meta-scientific perspective, the exploratory analyses illustrate how many-analysts simulation studies can reveal a broader space of plausible methodological decisions than is typically represented in a single simulation study. This broader design space may uncover sources of robustness and fragility that remain hidden when simulation settings are specified by only one research team.

Recommendations for the conduct of many-analysts simulation studies

In this section, we derive general recommendations on the conduct of many-analysts simulation studies. [TO BE UPDATED BASED ON THE RESULTS].

Regarding RDF in the analysis of the confirmatory part, the lead team can provide analysts with the results of the entire simulation study (covering the simulation results produced by the design choices of all included analysts) and ask them to select a suitable analysis strategy. This may involve making analytical choices to aggregate results across repetitions, potentially to obtain a ranking of the competing methods (Nießl et al., 2022), and choices to handle method failures (Pawel et al., 2025; Wunsch et al., 2025). RDF in the interpretation of simulation studies can be addressed by asking a subset of analysts to describe the results in a blinded fashion (in which the different methods are for instance labelled as method A, B, C, see (Boulesteix, Wilson, & Hapfelmeier, 2017)) to avoid result-dependent selective reporting and spin.

Author Contributions

Conflicts of interest

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Supplementary Material

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Table 1

Researcher degrees of freedom explored in the confirmatory part of the many-analysts simulation study. The table summarizes analysts' choices regarding simulation design characteristics. Weak, moderate, and strong conditions refer to pre-specified ranges of the corresponding parameter values.

Parameter	Characteristics / value type	Overall (N = XX)
<i>Categorical parameters</i>		
Distribution type	Parametric	X (XX.X)
	Real-sampling	X (XX.X)
X-type	Numeric	X (XX.X)
	Binary	X (XX.X)
Restriction type X	Deterministic	X (XX.X)
	Probabilistic	X (XX.X)
Restriction type Y	Deterministic	X (XX.X)
	Probabilistic	X (XX.X)
Use of covariates	Yes	X (XX.X)
	No	X (XX.X)
Covariates correlated with X	Yes	X (XX.X)
	No	X (XX.X)
<i>Continuous parameters</i>		Mean (SD)
Convenience-sample size		X (XX.X)
True R^2	All	X (XX.X)
	Weak	X (XX.X)
	Moderate	X (XX.X)
	Strong	X (XX.X)
Variance reduction X	All	X (XX.X)
	Weak	X (XX.X)
	Moderate	X (XX.X)
	Strong	X (XX.X)
Variance reduction Y	All	X (XX.X)
	Weak	X (XX.X)
	Moderate	X (XX.X)
	Strong	X (XX.X)

Table 2

Meta-regression models examining how relative bias varied as a function of researcher degrees of freedom in the simulated data-generating processes across effect size measures, subjective effect size categories, and restriction severity categories.

Variable	Category	Sample X			Sample Y		
		β_1	β_1^{std}	Cohen's d	β_1	β_1^{std}	Cohen's d
Intercept		XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
Restriction	Weak	0	0	0	0	0	0
	Moderate	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
	Strong	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
Effect Size	Weak	0	0	0	0	0	0
	Moderate	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
	Strong	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
Restriction mechanism	deterministic	0	0	0	0	0	0
	probabilistic	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
X-distribution	parametric	0	0	0	0	0	0
	real-world	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
Sample Size		XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
X Type	Binary	0	0	-	0	0	-
	Numeric	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	-	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	-
Covariates	No	0	0	0	0	0	0
	Yes	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]
Covariates correlated with X	No	0	0	0	0	0	0
	Yes	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]	XX.X [XX.X, XX.X]

Note: Sample Size refers to the size of the convenience samples. Because R^2 and Pearson's r are functionally redundant with the standardized regression coefficient, we did not conduct separate meta-regressions for these measures.

Figure 1. Grey points show the full sample; coloured points show the convenience sample (deterministic on the left, probabilistic on the right). Restricting X leaves the slope unbiased (up to sampling variability), whereas restricting Y leads to substantial deviations from the true slope.

Figure 2. Mapping between subjective effect size categories and their corresponding objective measures across different data-generating processes (DGPs). Each line represents a DGP, with points corresponding to weak, moderate, and strong effects (a) or restriction (b). Panel (a) shows the true partial R^2 across effect size categories. Panel (b) shows the variance reduction induced by different levels of restriction (e.g., range restriction) applied to X or Y; negative values indicate increased variance.

Figure 3. (a) Deterministic restriction ranges across DGPs. Intervals show the mean lower and upper percentile bounds of the restriction variable, for each restriction scenario, separately for restrictions based on X and Y.

(b) Probabilistic restriction based on Y. The x-axis shows the percentile of Y, and the y-axis shows the relative inclusion probability, normalised within each DGP and restriction category. Lines correspond to DGPs.

(c) Probabilistic restriction based on X. The x-axis shows percentiles of X, and the y-axis shows relative inclusion probability, normalised within each DGP and restriction category. Lines correspond to DGPs. Step-like patterns arise for binary X, with direction indicating whether higher or lower values are more likely to be retained.

Deterministic and probabilistic range restrictions across data-generating processes (DGPs) and restriction strengths. Inclusion probability is the probability that an observation is retained in the sample after the restriction is applied.

Figure 4. Aggregated relative bias results plotted against the true partial R^2 across restriction scenarios for restriction in X (upper panels) and Y (lower panels). Each analyst contributes multiple observations. Grey points and lines show each analyst's mean with its MCSE; coloured lines show LOESS smooths (span = 1) with its 95% confidence band, included

for descriptive purposes only. The dotted horizontal line marks zero relative bias.

Note. At 10,000 iterations, the MCSE is negligible and therefore not visible. LOESS confidence bands reflect uncertainty in the smoothed trend; between-analyst heterogeneity is examined separately in Supplement S2.

Figure 5. Non-aggregated power for the unstandardized regression coefficient β_1 by restriction strength. Black dots represent the power in the unrestricted sample, ordered from lowest to highest R^2 ; suffixes w , m , and s following the DGP number indicate the subjective effect size categories, respectively.

Figure 6. Prior and posterior beliefs regarding hypothesis plausibility. **H1.1** concerns the unstandardized regression coefficient, **H1.2** Cohen's d , and **H1.3** standardized regression coefficients, correlations, and R^2 under restriction in X. **H2** concerns restriction in Y. **H3** concerns power. Percentages indicate changes in beliefs, and line thickness reflects how many analysts changed in each direction.

Figure 7. Consensus classification for DGPs across hypotheses, effect sizes, and restriction strengths. Bars show the percentage of DGPs classified as supporting, contradicting, or yielding inconclusive evidence for each hypothesis. Panels are faceted by true effect size (rows) and restriction strength (columns). Classifications were based on equivalence testing using 95% Monte Carlo CIs around relative bias estimates, with a $\pm 5\%$ equivalence rule.